

Laura S. Bumbulis

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Education, University of Waterloo

PhD Statistics

2024–present

Advisor: Richard J. Cook
Overall Average: 95%

MMath Statistics

2023–2024

Masters Research Paper: *Robustness and Efficiency Considerations when Testing Process Reliability with a Limit of Detection*
Advisor: Richard J. Cook
Overall Average: 97%

BMath (Co-op)

2017–2022

Majors: Statistics; Combinatorics & Optimization
Minor: Pure Mathematics
Overall Average: 94%

Peer-Reviewed Publications

2. Colaco, K., **Bumbulis, L.**, Chandran, V., Cook, R. J., Gladman, D. D., & Eder, L. (2025). The Effect of Biologic Therapies on Serum Metabolic Biomarkers in Patients with Psoriatic Arthritis. *The Journal of Rheumatology*, 52 (Suppl 2), 27–28. [DOI:10.3899/jrheum.2025-0314](https://doi.org/10.3899/jrheum.2025-0314). [TOUR2C](#)
1. **Bumbulis, L. S.**, & Cook, R. J. (2025). Robustness and Efficiency Considerations when Testing Process Reliability with a Limit of Detection. *Mathematics*, 13 (8), 1274. [DOI:10.3390/math13081274](https://doi.org/10.3390/math13081274)

Research Funding

NSERC Canada Graduate Scholarship – Doctoral	2025–2028	\$ 120,000
Ontario Graduate Scholarship (OGS) / QEII-GSST (Declined)	2025–2026	\$ 15,000
University of Waterloo Graduate Research Studentship (×6)	2023–2025	\$ 74,000

Scholarships & Awards

Conference Presentation Awards

SSC Student Research Presentation Award: Biostatistics Competition	2025	\$ 500
Waterloo Student Conference in SAS & Finance Presentation Award	2024	\$ 500

University of Waterloo Scholarships & Awards

Doctoral

President's Graduate Scholarship	2025–2026	\$ 10,000
SAS Department Comprehensive Examination Scholarship	2025	\$ 1,000
Russell Hiscock Graduate Scholarship in SAS	2025	\$ 2,000
SAS Chair's Graduate Award	2025	\$ 500
University of Waterloo Graduate Scholarship	2025	\$ 1,000
SAS Department Teaching Assistant Award	2024	\$ 500
Cotton Family Women in Mathematics Graduate Scholarship	2024	\$ 3,000
Provost Doctoral Entrance Award for Women	2024	\$ 5,000

Masters

SAS Department Outstanding Academic Performance Award	2024	\$ 500
University of Waterloo Graduate Scholarship (×2)	2024	\$ 3,000
D.A. Sprott Entrance Award	2023	\$ 1,000

Undergraduate

President's International Experience Award	2022	\$ 1,500
Faculty of Mathematics Scholarship	2017–2022	\$ 10,000
Co-op Student of the Year, Faculty of Mathematics	2021	
U Sports Academic All-Canadian	2018–2021	
President's Research Award	2018	\$ 1,500
President's Scholarship of Distinction	2017	\$ 2,000

Presentations

Contributed Conference Presentations

3. *Dual Censoring Schemes for Marker and Failure Time Processes*. SSC Annual Meeting, Saskatoon, SK. May 2025.
2. *Robustness and Efficiency Considerations when Testing Process Reliability with a Limit of Detection*. Waterloo Student Conference in SAS & Finance, Waterloo, ON. October 2024.
1. *Testing Process Reliability under a Limit of Detection: Issues of Robustness and Efficiency*. SSC Annual Meeting, St. John's, NL. June 2024.

Work Presented by Co-Authors

2. Eder, L.*, Shahab, S., Gillespie, S., **Bumbulis, L.**, Emanoilidis, H., Compher, C., Scher, J., Gladman, D. D., Cook, R. J., Chandran, V., & Ogdie, A. *Dietary interventions in Psoriatic Arthritis: A Randomized controlled clinical trial*. ACR Convergence, Chicago, IL (USA). October 2025.
1. Colaco, K.*, **Bumbulis, L.**, Chandran, V., Cook, R. J., Gladman, D. D., & Eder, L. *The Effect of Biologic Therapies on Serum Metabolic Biomarkers in Patients with Psoriatic Arthritis*. ACR Convergence, Washington D.C. (USA). November 2024.

* Indicates presenting author.

Teaching Experience

Guest Lectures & Reviews

STAT 450: Estimation and Hypothesis Testing	Winter 2025
Instructor: Peter W. MacDonald	
Guest lecture on Bayesian estimation (50 mins)	
STAT 850: Estimation and Hypothesis Testing (Graduate-Level)	Fall 2024
Instructor: Yingli Qin	
Two final exam review sessions (3 hours each)	

Teaching Assistantships

STAT 231: Statistics	Spring 2025
STAT 230: Probability (Online)	Winter 2025
STAT 450: Estimation and Hypothesis Testing	Winter 2025
STAT 330: Mathematical Statistics [†]	Fall 2024

Credentials

Foundations for University Teaching in SAS Certificate	Winter 2025
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[†] Received SAS Department TA award.

All teaching experience affiliated with the University of Waterloo; see course descriptions [here](#).

Professional Activities

Co-organizer for SAS Student Seminar Series	2025–present
Graduate representative for SAS Department	2025–2027
Member of CANSSI Ontario Student PhD and Research Network (SPARK)	2025–2026
Organizer of 2026 CANSSI Ontario STatistics (CAST) seminar by Kuan Liu	
Graduate representative for Women in Mathematics Committee	2024–2025
Co-organizer for 2025 NYC Conference, University of Waterloo	
Co-organizer for 2025 OGMC, University of Toronto	
Session chair for Waterloo Student Conference in SAS & Finance	2024

All professional activities affiliated with the University of Waterloo.

Work Experience

Data Developer (Full-time), Emmetros	Jan.–Apr. 2023
Data Scientist (Co-op), Omnium	Jan.–Apr. 2021, May–Aug. 2022
Nominated for 2022 CEIA Student Achievement Award	
Bioinformatics Analyst (Co-op), Ontario Institute for Cancer Research	May–Aug. 2021
Awarded 2021 Co-op Student of the Year, Faculty of Mathematics	
Data Developer (Co-op/Part-time), Emmetros	Sep. 2019–Dec. 2020
Nominated for 2020 Co-op Student of the Year, Faculty of Mathematics	

Additional Activities

Intramural Sports , University of Waterloo	
Volleyball [‡]	Winter 2025
Flag Football	Spring 2024
Varsity Swim Team , University of Waterloo	
Volunteer Assistant Coach	2021–2022
Co-Captain	2020–2021
Member	2017–2021
Music	
Piano Player	2007–present
RCM Level 9 Harmony Certificate	2013
RCM Level 7 Piano Certificate	2013

[‡] *Indicates team captaincy.*

Acronyms

ACR	American College of Rheumatology
CANSSI	Canadian Statistical Sciences Institute
CEIA	Canadian Education & Internship Association
NSERC	Natural Sciences & Engineering Research Council
NYC	Navigating Your Careers
OGMC	Ontario Graduate Mathematics Conference
QEII-GSST	Queen Elizabeth II Graduate Scholarship in Science & Technology
RCM	Royal Conservatory of Music
SAS	Statistics & Actuarial Science
SSC	Statistical Society of Canada